Configuration Details:

System name of the submission: **MLFiles**
Submitter name: **Nutanix with Supermicro**
Submission date: **07/10/25**
Version of the benchmark: **2.0.0b1**
Solution type of the submission: **Shared File**
Submission division (OPEN or CLOSED): **CLOSED**
Power Requirements: **32600 Watts**

Host - Hypervisor

- AHV version 10.3
- 2x32 CPUs, Intel(R) Xeon(R) Gold 6448H @ 2.8 GHz
- 1024 GB DDR5 physical memory

CVM

- 32 vcpus
- 256GB RAM
- AOS version 7.3
- iSER data path enabled
- Compression disabled

FSVM

- 32 vcpus
- 200GB RAM (for Cosmoflow used 350GB)
- Files version 5.3
- SRIOV NIC for client traffic
- NFS over RDMA enabled
- RDMA max connection count 2048
- 2 FSVMs per host
- Performance profile
- Share compression disabled

Share and Data layout

- /root used general share
- Distributed shares submounted under /root:
    - Unet3d: /root/unet3d/train
    - Resnet50: /root/resnet50/train
    - Cosmoflow: /root/cosmoflow/train

- Top level directories:
    - Resnet50 and unet3d - 960 top level dirs
    - Cosmoflow - 1920 top level dirs
- 17 vdisks per FSVM for share
- Share workload type:
    - Cosmoflow - default share type
    - All other tests used sequential share type.

Clients

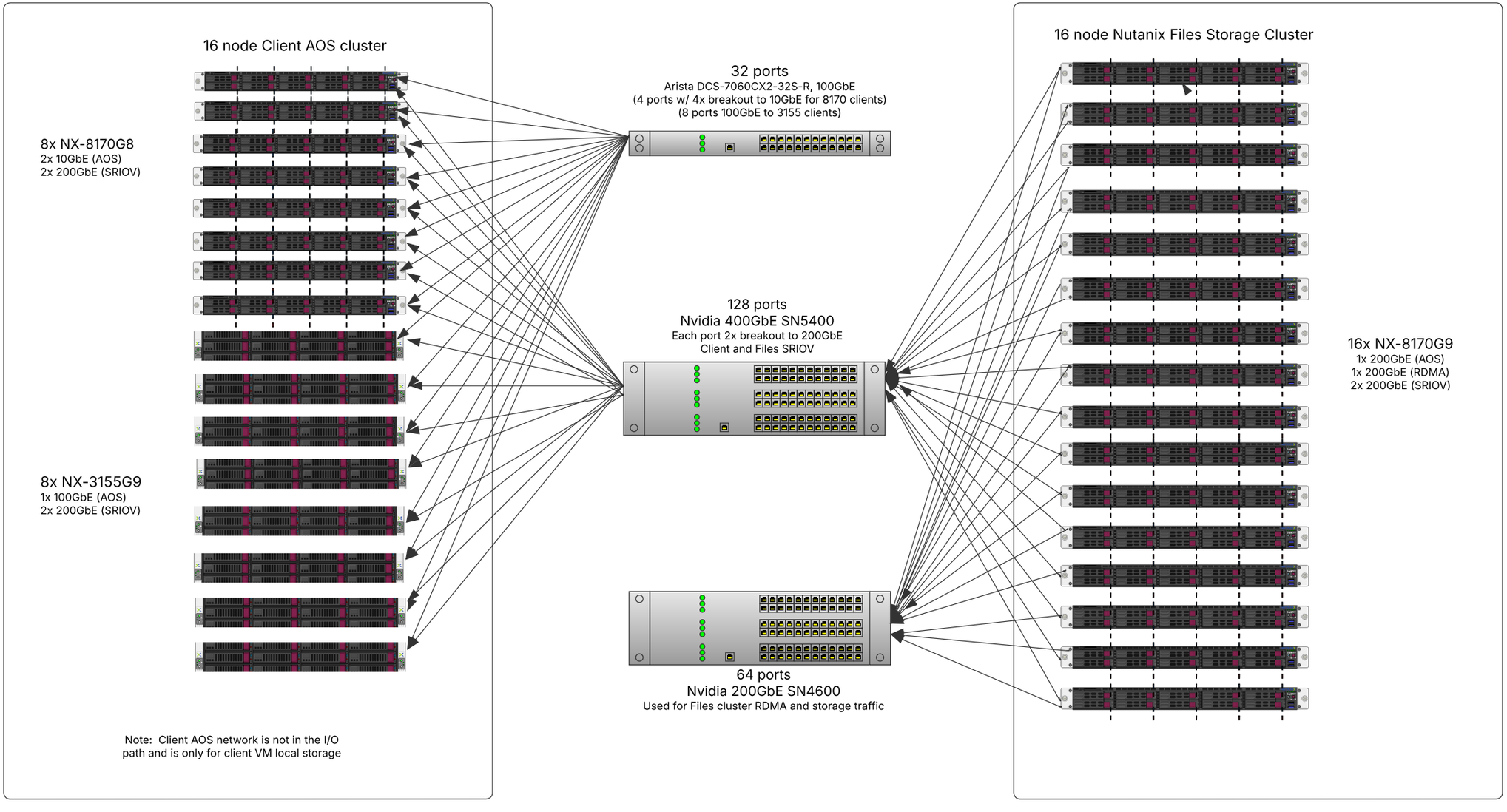
- Hosts - 8x NX-8170G8, 13x NX-3155G9 (note, some tests did not utilize all client hosts)
- VMs hosted on AHV 10.3, AOS 7.3 cluster
- 4x client VMs per host for G9 systems, 2x per host for G8.
- Ubuntu 24.04, kernel 6.9
- 8170-G8 hardware - SuperMicro, Intel Xeon Platinum 8352Y 2x32 CPUs
- 3155-G9 hardware - SuperMicro, Intel Xeon Gold 6548Y+, 2x32 CPUs
- 1024GB Physical RAM per host
- Client VMs
    - 32 vcpus
    - 64GB for cosmoflow
    - 192GB for resnet50
    - 192GB for unet3d
- Swap off
- NFS mount with RDMA, v4.0 and nconnect=4
- sunrpc.tcp_slot_table_entries=32
- NFS read_ahead - defaults
- Direct IO used for unet3d.

Example Benchmark Run Commands:

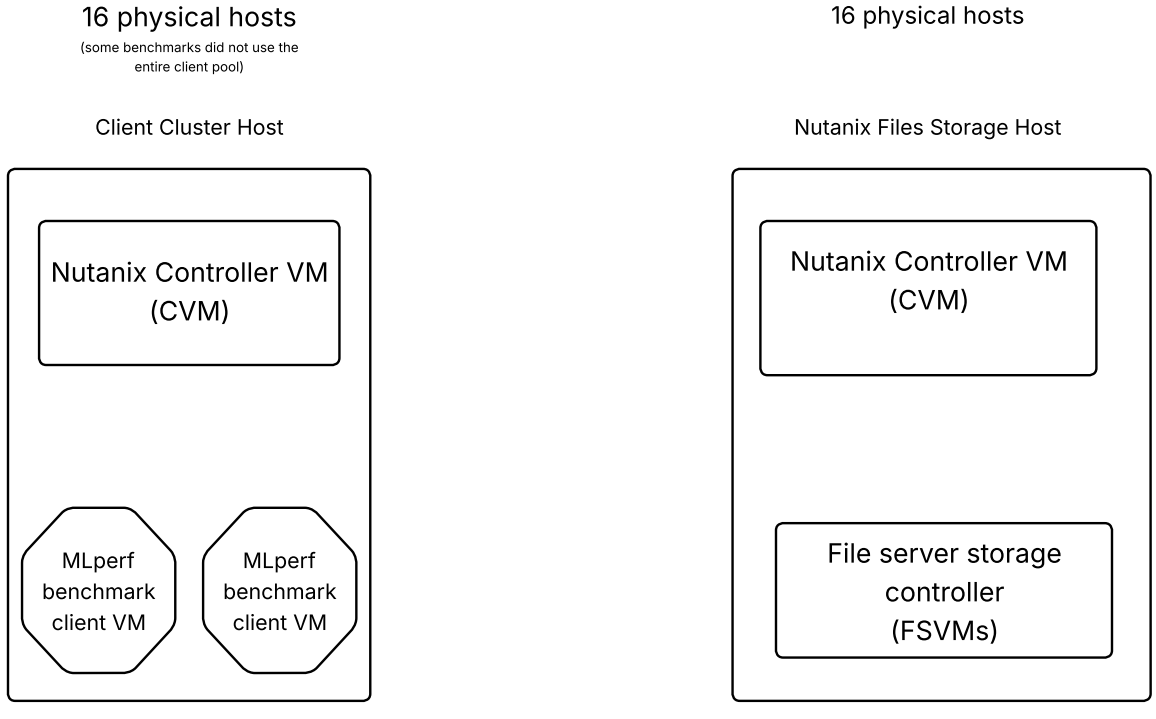
```
 mlpstorage training run --hosts
10.57.205.101,10.57.205.102,10.57.205.103,10.57.205.104,10.57.205.105,10.57.2
05.106,10.57.205.107,10.57.205.108,10.57.205.109,10.57.205.110,10.57.205.111,
10.57.205.112,10.57.205.113,10.57.205.114,10.57.205.115,10.57.205.116,10.57.2
05.117,10.57.205.118,10.57.205.119,10.57.205.120,10.57.205.121,10.57.205.122,
10.57.205.123,10.57.205.124,10.57.205.125,10.57.205.126,10. 57.205.127,10.57.2
05.128,10.57.205.129,10.57.205.130,10.57.205.131,10.57.205.132,10.57.205.133,
10.57.205.134,10.57.205.135,10.57.205.136,10.57.205.137,10.57.205.138,10.57.2
05.139,10.57.205.140,10.57.205.141,10.57.205.142,10.57.205.143,10.57.205.144,
10.57.205.145,10.57.205.146,10.57.205.147,10.57.205.148,10.57.205.149,10.57.2
05.150,10.57.205.151,10.57.205.152 --num-client-hosts 52
--client-host-memory-in-gb 64 --num-accelerators 52  --accelerator-type h100
--model unet3d --param dataset.num_files_train=20020 --data-dir /mnt/data
--results-dir unet3d_trial --allow-run-as-root --params
dataset.num_subfolders_train=960 reader.odirect=true --params
reader.read_threads=16
```

Example NFS mount command from Ubuntu client:

```
sudo mount -t nfs -o rdma -o vers=4.0 10.57.205.61:/resnet50 /mnt/resnet50
```


16 node Client AOS cluster
8x NX-8170G8
2x 10GbE (AOS)
2x 200GbE (SRIOV)
8x NX-3155G9
1x 100GbE (AOS)
2x 200GbE (SRIOV)
Note: Client AOS network is not in the I/O path and is only for client VM local storage
32 ports
Arista DCS-7060CX2-32S-R, 100GbE
(4 ports w/ 4x breakout to 10GbE for 8170 clients)
(8 ports 100GbE to 3155 clients)
128 ports
Nvidia 400GbE SN5400
Each port 2x breakout to 200GbE
Client and Files SRIOV
64 ports
Nvidia 200GbE SN4600
Used for Files cluster RDMA and storage traffic
16 node Nutanix Files Storage Cluster
16x NX-8170G9
1x 200GbE (AOS)
1x 200GbE (RDMA)
2x 200GbE (SRIOV)

Virtual environment for each host:



System Compon	ePower Supply Unit	Nameplate Rated Power(W)	Design Power(W) *	Rack Unit
SN4600 switch	Power Supply 1	600	600	2
	Power Supply 2	600		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1

	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1

	Power Supply 2	2000		

Total		65200	32600	18

*Note: All power supplies are redundant, so Design power is half of Nameplate power.